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<table>
<tr><td>LESSON PLAN</td><td>COURSE: MHF4U</td></tr>
<tr><td>Date: Lesson #9</td><td>Unit of Study:

- Chapter#4 Trigonometry
- Chapter #5 Trigonometric Functions</td></tr>
<tr><td>Materials / Resources:

- Advanced Function 12 Study Guide
- Chapter #4 & 5 Formula, Aid and Homework Sheet(s)</td><td>Lesson Focus

- Compound-Angle Formula
- Prove Trigonometric Identities
- Graphs of Sine, Cosine and Tangent Functions</td></tr>
<tr><td colspan="2">Prior Knowledge:

- Basic trigonometric knowledge
- Property of a triangle and their characteristics
- Basic trigonometric identities (e.g. $x \ + \ x \ = 1$)
- Able to describe key properties of periodic functions</td></tr>
<tr><td colspan="2">Overall Expectations:

1. Make connections between trigonometric ratios and the graphical and algebraic representations of the corresponding trigonometric functions and between trigonometric functions and their reciprocals, and use these connections to solve problems
2. Solve problems involving trigonometric equations and prove trigonometric identities</td></tr>
<tr><td colspan="2">Specific Expectations:

2.1 Sketch the graphs of $f(x) = \sin sin\ x$ and $f(x) = \cos cos\ x$ for angle measures expressed in radians, and determine and describe some key properties (e.g., period of 2π , amplitude of 1) in terms of radians
2.2 Make connections between the tangent ratio and the tangent function by using technology to graph the relationship between angles in radians and their tangent ratios and defining this relationship as the function $f(x) = \tan tan\ x$, and describe key properties of the tangent function
3.2 Explore the algebraic development of the compound angle formulas (e.g., verify the formulas in numerical examples, using technology; follow a demonstration of the algebraic development [student reproduction of the development of the general case is not required]), and use the formulas to determine exact values of trigonometric ratios [e.g., determining the exact value of $\sin sin \left(\frac{\pi}{12}\right)$ by first rewriting it in terms of special angles as $\sin sin \left(\frac{\pi}{4} - \frac{\pi}{6}\right)$]
3.3 Recognize that trigonometric identities are equations that are true for every value in the domain (i.e., a counter-example can be used to show that an equation is not an identity), prove trigonometric identities through the application of reasoning skills, using a variety of relationships (e.g., $\tan tan\ x \ = \frac{\sin sin\ x}{\cos cos\ x}$;</td></tr>
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$x + x = 1$; the reciprocal identities; the compound angle formulas), and verify identities using technology

Learning Goals:

- Students will be able to apply the compound-angle formula for sine and cosine
- Students will be able to solve problems involving compound –angle formulas
- Students will be able to prove trigonometric identities
- Students will be able to determine the characteristics of graphs of sine, cosine and tangent

Success Criteria:

1. By the end of this lesson I will be able to apply the compound-angle formulas for sine and cosine
2. By the end of this lesson I will be able to solve problems involving compound-angle formulas
3. By the end of this lesson I will be able to prove trigonometric identities
4. By the end of this lesson I will be able to determine the characteristics of graphs of sine, cosine and tangent

Assessment Strategies:

<u>Formative</u>: homework for chapters 4.4, 4.5 & 5.1

<u>Summative</u>: quiz #4, test #3

Time: **Part 1**	**Introduction:** Take up / check homework from lesson #8 and check for student's understanding. If needed, go over specific questions.
Time: **Part 2**	**Lesson:** <u>Chapter #4.4 Compound-Angle Formula</u> Given the proof of compound-angle formula, use examples from the text to solve for basic problems. <u>Chapter #4.5 Prove Trigonometric Identities</u> Starting off with reviving the most basic trigonometric identities (e.g. $x + x = 1$), review all formulas from this chapter and make a list for future references. Use a simple trigonometric equation as an example, start off by choosing either left side or right side to work from and from either one, start substituting terms and/or the equation with other trigonometric formulas obtained from this chapter. Continue to adjust the equation until it equals to the other side. <u>Chapter #5.1 Graphs of Sine, Cosine and Tangent Functions</u> With the aid from the formula sheets, define all characteristics of graphs of sine, cosine and tangent and then relate back to the function of these graphs. Relate graphs of sine and cosine to each other and let students recognize that they differ only from the placement of the graphs (e.g. cosine is just $\pi/2$"behind" sine). Also relating the graph

	of tangent to its alternative form, $\frac{\sin\sin x}{\cos\cos x}$, and recognize the reasoning behind tangent's asymptotes is because cosine cannot equal to zero, and therefore for all points that cosine does equal to zero (relating back to the cosine back, thus points $0,\ \pi,\ 2\pi, \ldots$ etc), there is a vertical asymptote (briefly review rational functions for reasoning)
Time: **Part 3**	**Wrap-up:** Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions.

Notes, reminders & homework:

Reminder:

- Quiz #4 (to be written before / after lesson #10)

Homework:

Chapter 4.4: # 1 – 10, 14, 18
Chapter 4.5: # 1 – 5, 10 – 12, 14
Chapter 5.1: # 1 – 12

Lesson Reflection: